



Institut luxembourgeois de la normalisation
de l'accréditation, de la sécurité et qualité
des produits et services

ILNAS-EN 15267-4:2017

**Air quality - Certification of automated measuring systems -
Part 4: Performance criteria and test procedures for
automated measuring systems for periodic measurements of
emissions from stationary sources**

Qualité de l'air - Certification des systèmes de mesure
automatisés - Partie 4 : Spécifications de performance et modes
opératoires d'essai des systèmes de mesure automatisés pour le
mesurage périodique des émissions de sources fixes

Luftbeschaffenheit - Zertifizierung von automatischen
Messeinrichtungen - Teil 4: Mindestanforderungen und
Prüfprozeduren für automatische Messeinrichtungen für
wiederkehrende Messungen von Emissionen aus stationären
Quellen

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National Foreword

This European Standard EN 15267-4:2017 was adopted as Luxembourgish Standard ILNAS-EN 15267-4:2017 in January 2017.

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**Air quality - Certification of automated measuring systems
- Part 4: Performance criteria and test procedures for
automated measuring systems for periodic measurements
of emissions from stationary sources**

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Prüfprozeduren für automatische Messeinrichtungen
für wiederkehrende Messungen von Emissionen aus
stationären Quellen

This European Standard was approved by CEN on 26 September 2016.

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COMITÉ EUROPÉEN DE NORMALISATION
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European foreword

This document (EN 15267-4:2017) has been prepared by Technical Committee CEN/TC 264 “Air quality”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by July 2017, and conflicting national standards shall be withdrawn at the latest by July 2017.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN [and/or CENELEC] shall not be held responsible for identifying any or all such patent rights.

This document is Part 4 of a series of European Standards:

- EN 15267-1, *Air quality — Certification of automated measuring systems — Part 1: General principles*
- EN 15267-2, *Air quality — Certification of automated measuring systems — Part 2: Initial assessment of the AMS manufacturer's quality management system and post certification surveillance for the manufacturing process*
- EN 15267-3, *Air quality — Certification of automated measuring systems — Part 3: Performance criteria and test procedures for automated measuring systems for monitoring emissions from stationary sources*
- EN 15267-4, *Air quality — Certification of automated measuring systems — Part 4: Performance criteria and test procedures for automated measuring systems for periodic measurements of emissions from stationary sources*

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, Former Yugoslav Republic of Macedonia, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey and the United Kingdom.

0 Introduction

0.1 General

CEN has established standards for the certification of automated measuring systems (AMS) used for monitoring emissions from stationary sources and ambient air quality. This product certification is based on the following four sequential stages:

- a) performance testing of the AMS;
- b) initial assessment of the AMS manufacturer's quality management system;
- c) certification of the AMS;
- d) post certification surveillance.

This European Standard specifies the performance criteria and test procedures for performance testing of portable automated measuring systems (P-AMS) used for periodic measurements of stationary source emissions. Testing applies to complete measuring systems.

NOTE 1 Portable electrical apparatus designed to measure combustion flue gas parameters of heating appliances are specified in EN 50379-1 to EN 50379-3.

The application of P-AMS for periodic measurements of stationary source emissions is based on

- specification of the standard reference method (SRM) and validation of the SRM;
- specification of the alternative method (AM) if the P-AMS is based on an AM;
- certification of the P-AMS in accordance with EN 15267-1, EN 15267-2 and EN 15267-4 including demonstration of equivalence with the SRM in the field if the P-AMS is based on an AM;
- on-going quality management by the user of the P-AMS in line with EN ISO/IEC 17025.

NOTE 2 Examples for standard reference methods for different measured components are listed in Annex F.

The overall assessment for the purposes of certification is *conformity testing*, while the evaluation of performance against specified performance criteria is *performance testing*.

0.2 Legal drivers

This European Standard supports the requirements of the following EU Directives:

- Directive 2010/75/EU on industrial emissions (integrated pollution prevention and control)
- Directive 2003/87/EC on processes emitting greenhouse gases.

However, this European Standard can also be applied to the monitoring requirements specified in other EU Directives.

0.3 Periodic measurements

Certified P-AMS can be used as SRM or AM for periodic measurements of stationary source emissions.

0.4 Relationship to EN 14181

Certified P-AMS can be used as SRM or AM for the calibration and validation of stationary AMS for QAL2 and AST purposes.

0.5 Processes

Field testing of P-AMS is ordinarily carried out on industrial processes representative of the range of application of the SRM or AM. The premise is that if the P-AMS performs acceptably on these processes, then experience has shown that the P-AMS generally performs well on the majority of other processes. However, there are always exceptions and it is the responsibility of the user to ensure that the P-AMS performs adequately on a specific process.

The necessary field test of P-AMS is specified in this European Standard.

0.6 Performance characteristics

A combination of laboratory and field tests is detailed within this European Standard. Laboratory testing is designed to assess whether a P-AMS can meet, under controlled conditions, the relevant performance criteria. Field testing, is designed to assess whether a P-AMS can continue to work and meet the relevant performance criteria in real applications including transportation to the measurement site, set-up of the P-AMS and measurement.

The main P-AMS performance characteristics are:

- response time;
- repeatability standard deviation;
- lack of fit (linearity);
- short-term drift;
- influence of ambient temperature;
- influence of voltage variations;
- influence of vibration;
- influence of sample gas pressure;
- influence of sample gas flow for extractive P-AMS;
- cross-sensitivity to likely interferents contained in the stack gas other than the measured component;
- converter efficiency for NO_x P-AMS;
- response factors for P-AMS measuring TOC;
- reproducibility under field conditions;
- trueness and precision of the P-AMS against the SRM under field conditions if the P-AMS is based on an AM.

Additional performance characteristics specific to the SRM or AM are included in the performance test.

The quality assurance and quality control (QA/QC) procedures to be applied by the user of the P-AMS are also assessed in the performance test.